

Digital Forensics: From Digital Evidence to Cyber Investigation

A technical walkthrough of how investigators identify, preserve, and analyze digital evidence to reconstruct real cyber incidents.



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CASE FILE

#DF-2026

STATUS

ACTIVE

SCOPE

IITDU

WHO AM I??

ZABER MAHMUD



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- Security Researcher | Blue Teamer | Penetration Tester
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What is Forensics?

The scientific process of examining evidence to establish facts admissible in a legal or professional context.



Purpose of Investigation

Apply structured, scientific methods to answer: what happened, when, how, and who was responsible — backed by verifiable evidence.



Core Principle

Findings must be objective, repeatable, and defensible — every conclusion traces back to preserved, untampered evidence.



Evidence

01

Raw material collected from the scene



Analysis

02

Systematic examination using proven methods



Findings

03

Objective observations drawn from analysis



Conclusion

04

Defensible answer to what actually happened

What is Digital Forensics?

The application of forensic science to digital devices and data — uncovering, preserving, and interpreting electronic evidence.



Digital Evidence

Any information of probative value stored or transmitted in binary form — files, logs, memory, network traffic, and more.

It must be handled so that it stays legally and technically trustworthy — this is why digital forensics follows a strict, repeatable process.

7

CORE STAGES

100%

CHAIN OF CUSTODY

0

TOLERANCE FOR
TAMPERING



Identification



Preservation



Acquisition



Examination



Analysis



Reporting



Documentation — recorded across every stage above

Forensics vs. Digital Forensics

Same investigative principles — applied to a fundamentally different kind of evidence.



Traditional Forensics

- **Physical evidence**

Fingerprints, fibers, blood, weapons

- **Fixed in space**

Tied to a physical crime scene

- **Visual examination**

Microscopes, chemical tests, ballistics

- **Degrades naturally**

Evidence can decay or be destroyed

VS



Digital Forensics

- **Digital evidence**

Files, logs, memory, network packets

- **Can exist anywhere**

Local disks, cloud, mobile, remote servers

- **Technical examination**

Hashing, imaging, specialized tooling

- **Fragile & volatile**

Can be altered or lost by a single reboot

Types of Digital Forensics

Ten specialized domains, each focused on a different source of digital evidence.



Computer / Disk

File systems & storage



Memory

Volatile RAM artifacts



Network

Traffic & packet capture



Mobile

Phones & tablets



Cloud

SaaS & remote storage



Database

Structured record stores



Email

Headers & message content



Browser / Web

History & web artifacts



Malware

Malicious code analysis

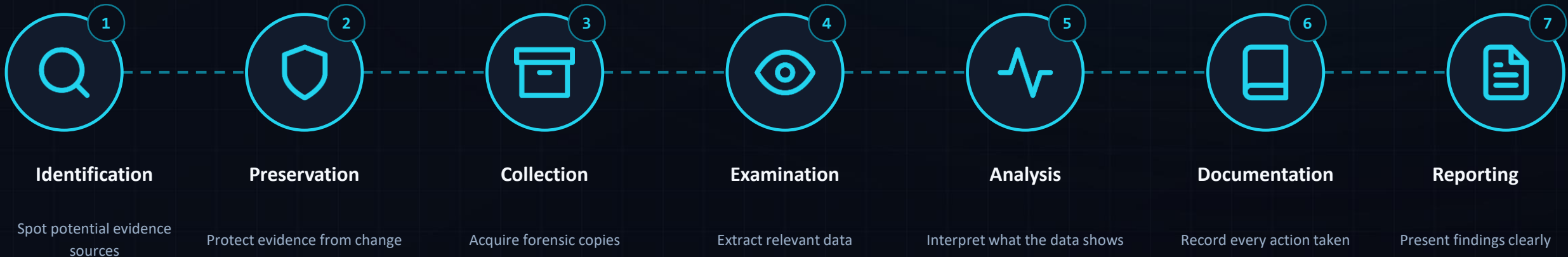


IoT / Embedded

Smart & embedded devices

Digital Forensics Investigation Model

A seven-stage chain of custody that keeps evidence sound from first contact to final report.



Chain of custody must remain unbroken across all seven stages — any gap can make evidence inadmissible.

Digital Evidence

What actually counts as evidence — anything that can help reconstruct events on a device or network.



Files

Documents, images, archives



Metadata

Hidden file properties



Logs

System & application events



Browser History

Sites visited & downloads



Emails

Headers, content, attachments



Network Packets

Captured traffic on the wire



Memory Artifacts

Running processes & keys in RAM



Deleted Files

Recoverable remnants on disk



Mobile Artifacts

App data, messages, GPS



Cloud Data

Synced files & account activity

Common Digital Forensics Formats

The container formats investigators encounter when imaging disks, capturing traffic, or dumping memory.

RAW / DD

Bit-for-bit disk image with no metadata — the universal baseline format.

E01 (EWF)

Compressed image with built-in hashing, used by EnCase & FTK.

AFF / AFF4

Open, extensible forensic container with embedded metadata.

VHD / VHDX

Microsoft virtual disk formats, common in Hyper-V environments.

VMDK

VMware virtual disk format encountered in server & cloud hosts.

QCOW2

QEMU/KVM virtual disk format used in Linux virtualization.

PCAP / PCAPNG

Captured network traffic for protocol & packet analysis.

Memory Dumps

Raw RAM capture (.mem/.raw) for volatile memory analysis.

ZIP / Archives

Compressed evidence bundles for storage & transfer.

Common Digital Forensics Tools

The core toolkit investigators reach for across disk, memory, network, and mobile analysis.



Autopsy

Disk & filesystem GUI



The Sleuth Kit

Disk forensics library



FTK Imager

Evidence acquisition



Magnet AXIOM

Multi-source artifacts



EnCase

Enterprise investigation



X-Ways Forensics

Disk & file analysis



Volatility

Memory forensics



Wireshark

Network / PCAP analysis



KAPE

Rapid artifact collection



Plaso / log2timeline

Timeline creation



Eric Zimmerman Tools

Windows artifact parsing



Cellebrite

Mobile device forensics



Ghidra

Reverse engineering



YARA

Malware & pattern ID

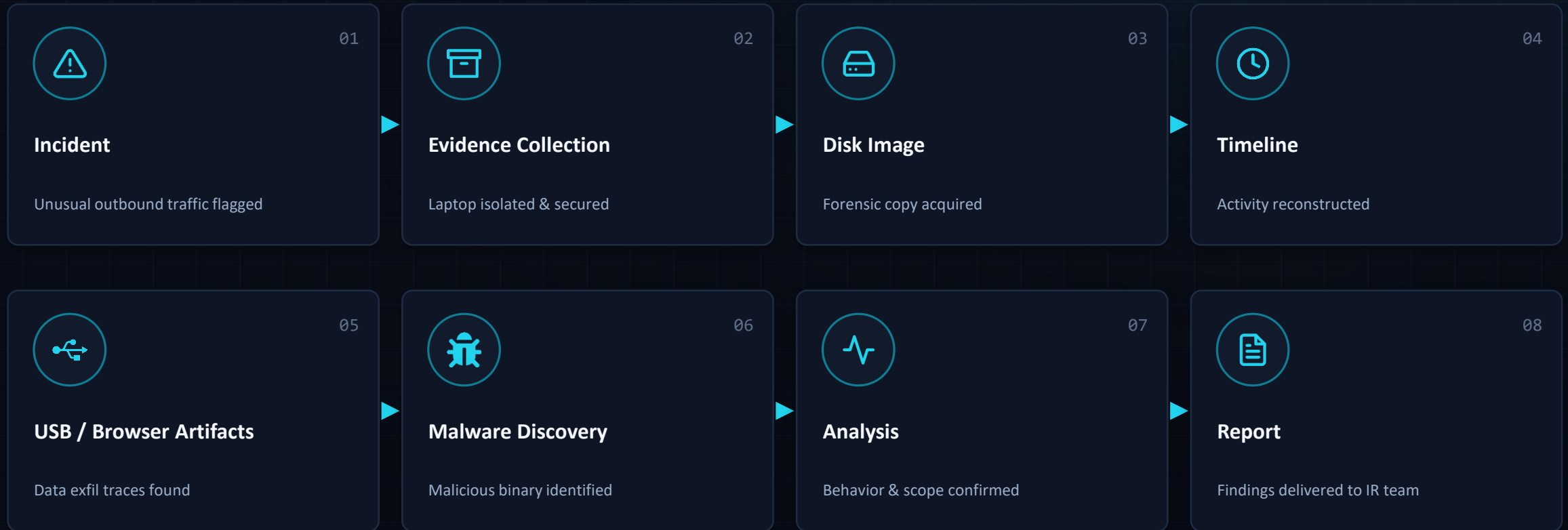
Most Common Software & Their Uses

A quick-reference matrix mapping each tool to its primary investigative role.

SOFTWARE	PRIMARY USE	SOFTWARE	PRIMARY USE
 Autopsy	Disk & filesystem investigation	 Wireshark	Network / PCAP analysis
 FTK Imager	Evidence acquisition & imaging	 KAPE	Rapid artifact collection
 Magnet AXIOM	Computer / mobile / cloud artifacts	 Plaso	Timeline creation
 EnCase	Enterprise forensic investigation	 Cellebrite	Mobile device forensics
 X-Ways	Disk and filesystem forensics	 Ghidra	Reverse engineering
 Volatility	Memory forensics	 YARA	Malware / artifact identification

Mini Investigation: Employee Laptop Compromise

A realistic, fictional incident walked through the full forensic pipeline — start to finish.



Where Digital Forensics is Used

From courtrooms to corporate SOC's — digital forensics underpins modern security and justice.



Cybercrime Investigation



Incident Response



Corporate Investigations



Financial Fraud



Insider Threats



Malware Investigations



Data Breach Investigations



Law Enforcement



E-Discovery



National Security

Real-Life Digital Forensics Jobs

Career roles across the DFIR ecosystem — from front-line SOC work to specialized examination.



Digital Forensic Analyst



DFIR Analyst



Incident Response Analyst



Malware Analyst



Memory Forensics Analyst



Mobile Forensic Examiner



Cybercrime Investigator



SOC Analyst → DFIR Pathway



Threat Hunter



Forensic Consultant



Digital Forensics Expert

Career Roadmap

A step-by-step path from foundational IT skills to a career in DFIR.



01

Networking + Linux + Windows



02

Cybersecurity Fundamentals



03

Digital Forensics Fundamentals



04

Disk / Memory / Network / Mobile Forensics



05

Hands-on Labs & CTFs



06

Certifications / Portfolio



07

DFIR / Digital Forensics Career

Skills You Need

The technical and analytical foundation every digital forensics investigator builds on.



Windows Internals



Linux



Networking



Filesystems



OS Artifacts



Python



PowerShell / Bash



Malware Fundamentals



Timeline Analysis



Log Analysis



Report Writing



Critical Thinking

How to Start Learning

A practical path from zero to your first real forensic investigation.



01

Build a forensic lab

Set up a safe, isolated practice environment



02

Analyze disk images

Practice with publicly available sample images



03

Practice memory forensics

Work through RAM dumps with Volatility



04

Analyze PCAP files

Study captured traffic in Wireshark



05

Investigate Windows artifacts

Explore registry, event logs, prefetch



06

Solve forensic CTF challenges

Apply skills under realistic pressure



07

Learn the core toolkit

Autopsy, FTK Imager, Volatility, Wireshark



08

Document your investigations

Build a portfolio of write-ups

“Digital Forensics is not just about finding files — it is about reconstructing what happened from digital evidence.”



Preserve Evidence

Protect integrity from first contact to final report



Analyze Systematically

Follow a repeatable, defensible process every time



Tell the Story with Evidence

Let the data — not assumptions — drive the conclusion



Questions?

Zaber Mahmud | EWU Cyber Security Club

IITDU SESSION • DIGITAL FORENSICS: FROM DIGITAL EVIDENCE TO CYBER INVESTIGATION

References & Further Reading

Authoritative sources for continued study in digital forensics and DFIR.

- NIST — Digital Forensics publications (Computer Forensics Tool Testing Program)
- NIST Computer Security Resource Center (csrc.nist.gov)
- SWGDE — Scientific Working Group on Digital Evidence
- The Sleuth Kit & Autopsy official documentation
- Volatility Foundation — memory forensics documentation
- Wireshark official documentation & user guide
- Magnet Forensics — knowledge base & resources
- Cellebrite — mobile forensics resources
- CISA — Cybersecurity & Infrastructure Security Agency resources